INRODUCTION

Scribe is a WikiMedia project that makes Wikidata based mobile keyboard applications to help second language learners communicate more confidently in the languages they're learning.

Many language learners rely on apps to learn vocabulary, grammar, and rules. However, when it comes to actually using the language, especially while writing messages, they are often on their own.

With Scribe, users can

- Look up plural forms
- Check translations
- Get help with verb conjugation and correct word forms

All of this happens without leaving the keyboard, reducing cognitive load and making language use more natural in everyday communication.

MY ROLE

I was responsible for designing and integrating a guess functionality into the existing user flow to encourage active recall during typing.

WHY THE GUESS FUNCTIONALITY

Scribe gave users the right answer, but nothing encouraged them to engage with the language themselves.

The guess functionality was introduced to address this gap. The intent was to encourage users to think and attempt an answer before receiving help, reinforcing learning through active recall while still preserving the convenience of in-keyboard assistance.

WHAT IS THE GUESS FUNCTIONALITY

The guess functionality (when enabled) prompts users to make a guess before the correct answer is revealed.

The feature was designed to be optional, allowing users to turn it off if they preferred faster assistance or did not want the additional interaction.

EXPLORING THE APPROACH

Once we knew we wanted a guess step, the open question was how much friction that step could add before it started working against the convenience Scribe is built on.

One of the first things I had to figure out was how the guess itself would be entered. I considered two directions: letting users select from a set of options, or having them type their answer freely. The difference mattered more than it might seem. Selecting from options relies on recognition, where the correct answer just needs to look familiar, while typing requires actually producing the answer from memory.

Cognitive psychology research on the "testing effect" shows that this kind of active retrieval leads to stronger long-term retention than recognition-based recall. Since the whole point of the guess step was to reinforce learning, not just add an interaction, typed input was the clearer choice.

The next question was how many attempts a user should get before the correct answer was revealed. I considered allowing multiple guesses, like a game, but that risked undermining the convenience Scribe is built around.

Users are typing to communicate, not to solve a puzzle, and repeated attempts could turn a quick lookup into a longer, more effortful detour. The PM agreed, and we settled on a single guess, enough to prompt genuine recall in the moment.

PROTOTYPE

1. Refining The Feedback Loop

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With the guess step now inline, the next question was how to surface whether a guess was correct. In my first iteration, the guess box itself showed the correct answer along with a tick or cross emoji as an interim confirmation, before the answer then appeared in the text area once the user pressed ‘Enter’.

The PM pointed out that this interim step was not necessary and suggested merging the checkmark directly onto the pill instead.

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In the next iteration, I removed the interim step. Pressing Enter now revealed the answer directly in the text area, with a tick merged onto the pill itself.

This introduced a new problem, though. Since the pill's background color varies based on grammatical gender, an emoji-based tick didn't have consistent contrast or legibility across all color combinations, raising accessibility concerns.

<insert image>

To resolve this, I replaced the emoji with a dedicated icon, ensuring clear, consistent feedback regardless of the pill's background color.

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2. Feasibility VS. Usability

The translation guess functionality required presenting users with multiple word senses, which are often lengthy. From a usability perspective, a scroll-based interaction would have made it easier to read and compare these longer definitions within the limited keyboard space.

However, the conjugation command already uses a side-click interaction to select the required verb form. To minimize additional development effort, I was instructed to reuse this existing pattern for selecting word senses in the translation flow.

While scrolling may have better supported reading longer definitions, reusing the existing side-click pattern prioritized feasibility within the project constraints.

FINAL FLOW

REFLECTION

At the time of writing, this feature has not shipped, as the team's priorities have since shifted to other areas of the product.

It also wasn't tested with real users before that shift happened, so while the design decisions throughout this project were grounded in reasoning, PM feedback, and engineering constraints, I don't have direct evidence of how users would actually respond to it in practice. If I were to continue this work, usability testing would be the natural next step, particularly around whether the guess step actually increases retention or simply adds friction.

If I were to do this again, I'd want to bring in user testing much earlier, even informal feedback on paper sketches like the pop-up concept, rather than relying only on internal feedback from the PM and engineering constraints. Some questions, like whether a single guess feels sufficient or whether typed input actually feels more engaging than selecting from options, can ultimately only be answered by real users.